Soft sensor: terminal blending RVP from component flows and assays

Goal. At an NGL/LPG blending terminal, continuously infer the blended tank's RVP-like volatility from each incoming stream's *periodic* lab assay and *continuous* flow rate — without an online analyzer on the blended output.

Terminals routinely blend several incoming NGL streams (e.g. a propane-rich stream, a butane-rich stream, a natural-gasoline/pentanes-plus stream) into a single blended product tank. Custody-transfer flow meters on each incoming line are cheap, continuous, and already there for inventory/billing. Each stream's composition, by contrast, usually comes from an infrequent lab GC sample — it is *not* continuously measured. A soft sensor can still track the blended product's volatility continuously by combining the last known per-stream assays with the live flow ratios, instead of waiting on (or paying for) a continuous analyzer on the blend itself.

Measurements assumed available: each incoming stream's composition (z_a, zb, zc — periodic lab assay, held constant between updates) and each stream's volumetric flow rate (continuous, from custody-transfer meters), plus blend temperature (continuous). **Inferred / computed by this soft sensor:** the blended stream's composition (flow-weighted, on a molar basis) and its bubble-point pressure (RVP proxy). **Tier:** A — direct calculation, given that each input stream's composition is known (even if only periodically) (`docs/measurement_tiers.md`). If even the per-stream assays become unavailable, blending ratios would have to be inferred from historical correlations instead — Tier C.

This report reuses the same Clapeyron.jl Peng-Robinson bubble-point calculation as `01_lpg_rvp_soft_sensor.jl`, applied to the *blended* composition rather than a single measured stream. The one new piece of physics: flow meters read volume, not moles, so each stream's volumetric flow has to be converted to a molar flow (via its liquid molar volume from the EOS) before the streams can be combined into a blended mole fraction.

> Teaching/demo report with representative, not terminal-specific, data, evaluated at a fixed operating point (the interactive notebook version lets you adjust flows and temperature with sliders).

1. Incoming stream assays (periodic lab data, held fixed here)

Component	Stream A (propane-rich)	Stream B (butane-rich)	Stream C (natural gasoline)
Propane	0.85	0.05	0.02
Butane	0.1	0.7	0.08
Isobutane	0.04	0.2	0.1
Pentane	0.01	0.05	0.8

2. Operating point used in this printout

Flows: Stream A 40.0 m³/h, Stream B 35.0 m³/h, Stream C 15.0 m³/h. Blend temperature: **25.0 °C**, pressure: **10.0 bar**.

Molar flows: Stream A ≈ 453.4 kmol/h, Stream B ≈ 358.3 kmol/h, Stream C ≈ 137.2 kmol/h.

```
v_A = volume(model, Pblend, Tblend, z_A; phase=:liquid)
n_A = q_A / v_A   # volumetric flow -> molar flow
```

3. Soft sensor output: blended composition & RVP proxy

Blended composition (mol fractions): propane 0.428, butane 0.324, isobutane 0.109, pentane 0.139.

Inferred blend bubble-point pressure (RVP proxy, soft-sensor output) at 25.0 °C: 512.1 kPa (74.3 psi)

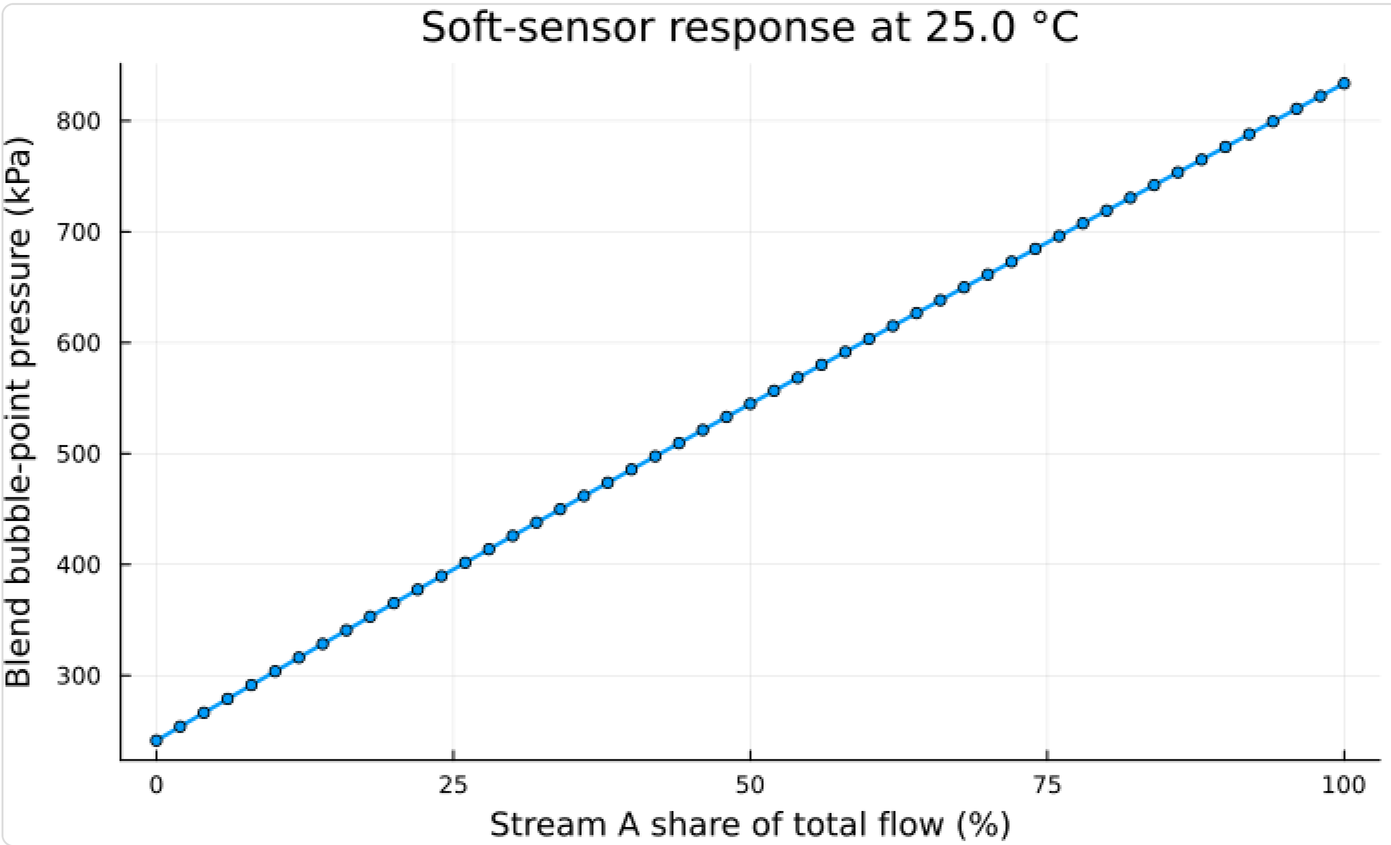
No analyzer touched the blended tank to get this number — it comes entirely from the three (periodic) stream assays and the three (continuous) flow meters.

```
n_component = n_A .* z_A .+ n_B .* z_B .+ n_C .* z_C
z_blend = n_component ./ sum(n_component)
p_bubble, vl, vv, y = bubble_pressure(model, Tblend, z_blend)
```

> Blend bubble-point pressure ≈ 512.1 kPa (74.3 psi) at 25.0 °C

5. Sensitivity: blend RVP proxy vs Stream A flow share

Sweep Stream A's flow (renormalizing B and C proportionally to keep total throughput fixed) at fixed temperature, to see how the blend's inferred RVP proxy responds to shifting more of the light, propane-rich stream into the blend. A soft sensor should move smoothly and monotonically here — more of the volatile stream should raise the blend's bubble-point pressure.



6. Next steps toward a deployable soft sensor

- Replace the fixed stream assays with values pulled from the plant historian / LIMS, refreshed whenever a new lab sample lands, while flow rate and temperature update every scan.
- Validate the inferred blend RVP against periodic tank-sample lab results; add a bias/trim term if a consistent offset appears.
- Track how much the inferred blend drifts from reality as time since the last per-stream assay grows — that bounds how long this Tier A treatment stays valid before a stream's composition should be treated as effectively unmeasured (Tier B/C) and estimated some other way (e.g. from upstream unit conditions).

- Extend to a live mass/energy balance across the blend tank (holdup, fill/draw dynamics) rather than an instantaneous flow-ratio blend, for terminals with significant tank residence time.

Generated from notebooks/03_terminal_blending_soft_sensor.jl (Clapeyron.jl PR EOS) — caxelrud/Soft-sensor